



GE Medical Systems

Technical Publication

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Revision 6

LightSpeed QX/i

Application Tips and Work-Arounds

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CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

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REVISION HISTORY

Revision	Date	Reason for change
0	11/6/98	3.1HM-2.5.4m Pilot Release
1	5/7/99	4.1_2.7.3gpc2 LightSpeed 1.0 Full Production Release
2	11/15/99	6.8.2_2.7.3n2 LightSpeed 1.1 Full Production Release
3	8/02/00	8.12.2_2.8.2G_H1.2M4 LightSpeed 1.2 Full Product Release
4	10/30/00	10.4.1_2.8.2I_H1.3M3a LightSpeed 1.3 Full Product Release
5	12/11/00	10.5_2.8.2I_H1.3M4 LightSpeed 1.3 Full Production Release
6	5/10/01	10.5_2.8.2I_H1.3M4 LightSpeed + FlexTrial Patch Release

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FEATURES VERSION 10.5_2.8.2I_H1.3.1M4+PATCH SOFTWARE

FLEX TRIAL

Flex Trial – The capability to Flex Trial software options through initialization via the Web is now possible. The following software features are available for Flex Trial: 3D Analysis, 3D Navigator, DentaScan, CT Perfusion and SmartPrep.

The update software to enable Flex Trial capabilities will be installed on your system by either an auto download via modem or by CD-Rom by your service engineer.

For sites receiving software by auto download, you will see two messages the next time you reboot your system and the software is available on the scanner:

1. “ Helios 1.3 patch successfully downloaded.” Select OK to continue.
2. “QX/i software update is available for automatic installation. Auto install will install the patch software, shutdown the system, and automatically reboot the system. Entire process is approximately 7-8 minutes.”

Select YES to automatically complete the download of the software. Select NO to cancel the download and a service engineer will be dispatched to install the update software.

To find out information about eFlexTrial visit CT Flextrial on the Web at gemedicalsystems.com under Radiology in CT under Special Promotions.

SCANRX

ScanRx – A fix is included to reduce incidence of a ScanRx crash occurring after a Confirm or Resume.

SMARTTREND

SmartTrend – A fix is included which provides comparison of FastCal files to improve the detectability of obstruction of the beam path due to foreign matter on the detectors, window of the tube of copper filter.

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PRECAUTIONS VERSION 10.5_2.8.2I_H1.3M4+PATCH SOFTWARE

SYSTEM

- Run Shutdown and Restart daily.
- The system may fail to start if system disk space needs to be recovered from unused system files. The following message will be posted in the Disk Management shell window:

```
"The purpose of storelog is to recover system disk space by archiving and then removing core, log, and data files that have been saved for their system troubleshooting diagnostic value. Now they may be taking up too much space for the system to run properly. Removing these "system log" files does not add image space, but should allow the application to start."
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You are then directed to answer the following questions (be patient for the messages to display):

Place the mouse cursor inside the black Disk Management window

Q: Do you want to save system log files to removable media

Type n and hit the enter key

Q: Do you want to remove the system log files

Type Y and hit the enter key.

You will then see a message "Please be patient, the system is now removing unwanted files." The system will then start normally.
- If the system fails to startup completely, select [Unix Shell] from the tool chest menu in the upper right of the Scan Monitor place your cursor in the blue shell and type st.
- If mouse selections fail, select the [Escape] key to clear backlogged requests.
- The cursor may not move between the left and right screen. Wait a few seconds without moving the cursor and try again.
- To install a SMPTE pattern, BRH or Quality assurance images for viewing, select the following. Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. Both will be listed as Exam 1000, the patient name will reflect if it is a QA image or a SMPTE pattern.
 - Select Service Desktop
 - Select [Diagnostics]
 - Select [Display Processing]
 - Select [Install SMPTE Image]
- In general, wait for a screen transition to take place before making another selection.
- If the console becomes unresponsive for 2 minutes or more, shutdown the system using the pink shutdown button and restart the system. If you cannot select the pink shutdown button, turn off the console power switch, wait 10 seconds, then turn the console power switch back on. The system should come up normally.
- The system will lock up if the mouse becomes hung on the left monitor. The only known recovery is to cycle power to the console.

SYSTEM CONTINUED

- You may see a pink warning dialog box posted indicating the system has low disk space. This is due to a partition on the system disk getting full. This can be caused by archive, network, or film queues containing too many entries. Make sure that the entries are processed. If the queue is paused, resume the queue or delete entries as needed.
- Protocol Management may not be selectable. Check that Retro Recon is selectable. If Retro Recon can not be selected, the system will need to be rebooted.
- If you are unable to type in any field in New Patient, or only field accepting input is the History field, move cursor to the field you want to type in and click the middle mouse. This should allow you to make entries. If this fails, a system shutdown is required.

TUBE WARM UP/FAST CAL

- The detector temperature has a 15 minute stabilization time immediately after a system restart. If Fast Cal is entered during this stabilization period a dialog will be posted instructing the user to wait x minutes before beginning Fast Cal.
Either wait or run Fast Cal later. Do not forget to run Fast Cal daily
- To avoid having to wait to run Fast Cal do one of the following:
 1. Run Fast Cal before performing Shutdown/Startup
 - or
 2. Perform Startup/Shutdown and Fast Cal separately
- Run Tube Warm up after 2 hrs. of non-use. It is best to warm the tube as close as possible to the time the next patient will be scanned. Don't warm the x-ray tube and then wait 30 minutes without scanning.
- Always check that the beam is clear when doing Fast Cal. In Fast Cal, the system checks cleanliness of the Mylar Window. If the system suspects the Mylar Window may be dirty to the point that it may cause a beam obstruction, a pop-up message will be displayed. The operator will be instructed to clean the Mylar Window and select [Retry].
- Fast Cal must be run once every 24 hours.
- Complete all portions of Fast Cal, Warm Up 1, Warm Up 2, Gen Cal, Clever Gain and Fast Cal. This assures that the Air calibration and generator calibrations are up to date on the system.

SCAN

- Scan aborts may occur during Axial or Helical scanning. Always be aware of the scan progress during an Exam and select Resume as soon as it is posted to continue.
- Scan may fail to confirm posting a message that not enough image space exists, even though the image space shown in the Feature Status Area indicates there is enough space. This is due to the fact that images are stored on the system disk in more than one partition. Remove consecutive exams to free up image space for confirm to proceed.
- To terminate an Insite connection on your system. Select [New Patient], a message will be posted informing you that the scan hardware resource is not available. Wait 3 minutes and select [New Patient] again to begin scanning.

SCAN CONTINUED

- In general, if a scan fails and a [Resume] is posted, press [Resume] to continue. Try [Resume] again if the first [Resume] fails. If a failure still occurs, reset the scanning hardware through System Resets in the Service desktop. If scan still fails to restart, shutdown and Restart the system.
- Auto Voice may fail to function, especially during system simultaneity. Make sure that you can hear the Auto Voice to recognize if Auto Voice has quit, manually breathe the patient when this occurs.
- It's best to use the Pause hard key button on the key board if pause during scanning is desired.
- The Show Localizer scout image may fail to display or images may fail to recon and may become suspended if scanning is started while remove images is in progress. To avoid this **don't** remove images while an exam is in progress or scanning is active. Remove images at system idle times.
- The Dynaplan screen may report incorrect status (i.e. a scan is removed before it is actually scanned) when stop scan is selected. The screen will reset correctly after Resume or back to view edit is selected.
- The ISD will not default to 1 sec. when a zero interval is selected for axial scans. Edit the ISD field to use a 1-sec. ISD.
- Scan groups will not be contiguous if you switch from 2 or more Helical scan groups column or row edit and the scan type is changed to Axial. If this change is made, then verify the Start and End locations of each of your scan groups and adjust if needed.
- The Cine time between images may change when other parameters are modified within a protocol. Check the Cine time between images prior to confirming the scan to verify it is the value you wish.
- Tilt handles will not be visible on the screen if the DFOV is larger than 48. Use a DFOV smaller than 48 to assure that the tilt handles are visible.
- Show Localizer may fail to display the scout image if the Next Series is selected before the scout images are reconstructed. Wait until the scout images are reconstructed before selecting Next Series. If Next Series has already been selected, toggle Show Localizer off and then on again.
- Quickly changing desktops when Show Localizer has been selected for the next series can cause a partial display of the scout in the Graphic Rx window to occur.
- If Page Up or Page Down fails to change the displayed Show Localizer.
Try the following in order:
 - Turn off then on the Show Localizer button
 - Move the cursor off the image window and back on to the image to refresh the screenIf both of these fail then verify that you have a valid scout for Show Localizer.
- When changing from a 3.75mm/11.25 HQ scan to HS mode the slice thickness will change to 5mm/22.5 with no notice to the operator. The 3.75 thickness is not valid at 22.5 table speed. If 3.75mm thickness is desired, then change your table speed.
- To copy Start/End location, Number of Images, Thickness, Interval, DFOV, R-L center and A-P center from Recon 1 to Recon 2 or 3 toggle Recon 2 or 3 from N to Y. This will need to be checked anytime you change the Start or End for Recon 1 when Recon 2 or 3 is turned on.
- If a R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.

SCAN CONTINUED

- A protocol may fail to display. The following message will be posted to the message line on the lower center of the scan monitor "Can't read selected protocol, please choose another protocol". If this occurs the protocol is corrupt and will need to be rebuilt. Delete the protocol and rebuild it.
- The patient history field may be missing the last 9 characters when all 60 characters have been entered and the patient record is either selected as a completed exam or the Patient ID is entered and the system matches a completed exam.
- Even weights entered in Patient Schedule may be rounded up to the next odd pound. This occurs because weights are stored as kilogram units then converted back to pounds.
- During the acquisition of large data sets (1000-1500), there may be Autofilm and reconstruction performance issues. If either stops, check the queues and restart to continue.
- When entering birthdates on the New Patient or Schedule Patient screens the following rules apply:
 - 2 digit years can only be entered if the birth year is in the current century for the 21st century - for the year 2000 forward.
 - Birthdates from any other century must be entered as 4 digits for example for the year 1899 – 1999 all four digits must be entered.
 - Birthdates can only be entered for 120 years (Current year minus 120 years).
- If InSite is running the "Remote Safety" test, New Patient will not open. New Patient can be made available by either:
 1. Calling InSite and request they abort the test.
 2. Going to the Service Desktop and selecting [CleanUp] option to cancel the test in progress.
- Please double check the Prep Delay entered on the Review Dynaplan Screen before scanning if it was modified from the what was originally defined in the protocol. It has been reported to have changed from what was manually entered. Return to the View/Edit screen and prescribe the desired value.
- A scan abort will occur if a "Buffer Overrun" is detected. Select the [Resume] key to restart the scan.
- PMR images may fail to recon if disk space is low. To recon the images, remove images and restart the recon queue in Recon Management.
- The scan range for Recon 2 and 3 for a 2i 0.63mm (sub mm) series is not adjustable. It is locked to the range of Recon1. This issue will be resolved in the next release of software.
- Recon 2 and 3 may fail to queue for recon for 0.63 images when Recon 1 contains 2 images. This caused by changing to 0.63mm thickness from any other 2i slice thickness and Recon 2 and 3 not updating the scan locations correctly. To avoid this, if you change the slice thickness to 0.63mm/2i, go to Recon 2 and 3 turn off then turn on reconenable to correctly update the start, end and number of images.
- A zero interval axial series if paused will add another group equal to the remaining number of images. Make to return to the View/Edit screen to delete this group to avoid acquiring additional images.

SMARTPREP (PURCHASED OPTION)

- Monitor phase scanning will abort if the Monitor ISD is set to less than 3 seconds when the rotation time used is 0.8 sec. Set the Monitor ISD to 3 seconds or greater to avoid this problem.
If this has occurred, do one of the following:
 - Press Monitor Phase to continue monitor scans.
 - Use the graphed data and the clock countdown on the SmartPrep display to determine when to select Scan Phase.
- The film composers will flash and cannot be hidden on the SmartPrep display. This will occur if the Auto Film or Manual Film composer buttons are selected after confirm has been selected for SmartPrep. Don't select the film composer buttons while the SmartPrep display is first coming up or is visible to avoid this.
- If reconstruction fails during SmartPrep, scanning will pause and post a message. Press [Monitor Phase] to resume scanning for the monitor phase.

PATIENT SCHEDULE

- Close the preferences screen before switching between Patient Schedule and New Patient.
- The Patient Name, Patient ID, Accession Number and Requested Procedure ID can only be edited if "Allow to Edit MWL" is set to yes in the Preference window.

CONNECTPRO (PURCHASED OPTION)

- Bar Codes will fail to be read by the Bar Code reader if the HSI/RIS system the barcode was created on has a different language keyboard than the CT system. For example, if your CT system has a French language keyboard then your HIS/RIS must have a French language keyboard. If it is not possible to have the same language keyboard on each system, then manually enter the Accession or Patient ID number or select the desired patient from the Patient Schedule list to display the patient information on the New Patient screen.
- Exams created locally with multiple Requested Procedure IDs will fail to restore from MOD. Make sure exams with multiple Requested Procedure IDs are sent from the HIS/RIS to the Modality Work List and not created on the scanner.

PERFORMED PROCEDURE STEP (PART OF CONNECTPRO OPTION)

- PPS will post the message "Failed to Start" if the remote server is down.
- Scouts and Screen Save images created in ExamRx or in the Viewer on the Image Works desktop are not PPS aware in this release of software.

RECONSTRUCTION

- If images fail to recon the following should be used in order:
 - Select [Recon Management], [Unsuspend Queue]
 If the image still fails to recon
 - Select [Recon Management], [Restart Queue]
 If the image still fails to recon
 - Select [Service Desktop], [System Resets], [Recon Reset]
 If this still fails
 - Shutdown and reboot the system.
- If a R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.
- Lung Algorithm
 - Provides edge enhancement between structures with large density differences, such as calcium and air.
 - Enhances the contrast of small objects. For best viewing and film quality, select a window width of 1000 to 1500 and a window level of -500 to -600.
 - Increases CT number values at the edge of high contrast objects. When planning to take CT number measurements of vessels or nodules in the lung, please check and compare your results with Standard algorithm images. (ROI and Histogram functions use CT numbers.)
 - The edge enhancement provided by Lung Algorithm may not be appropriate in some clinical cases. Please take individual viewing preferences into account when you choose Lung algorithm.
- Increased beam hardening artifact may be seen when imaging the posterior fossa on some patients and utilizing 120kVp. Use 140 kVp through the posterior fossa area to avoid this.
- Images may fail to reconstruct with a bad syntax error logged in the error log. This occurs when the name of the protocol used for scanning has a Control M or / in the protocol. Do not use these keys when naming a protocol to avoid this problem.
- Reconstruction may initially fail to reconstruct when the interval for helical scans is 2 or more times greater than the slice thickness for HQ mode helical scans. The system will automatically retry to complete the reconstruct but the recon time per image will be increased. The following table indicated the maximum interval allowed for the following slice thicknesses and table speed for HQ mode:

Slice Thickness	Table Speed	Maximum Interval
2.5	3.75	4.2
5	7.5	8.5
7.5	11.25	12.5
10	15.0	17.5

RETRO RECON

- Two decimal points cannot be entered, i.e. 3.75 for image interval if column edit is used. Edit each scan group (row) if a interval with 2 decimals is required.
- If a 1 rotation 4i or 2i axial scan is prescribed using add group in Viewedit with a Superior to Inferior scan direction is retro reconned, the resulting images will be reconned Inferior to Superior. The image locations are correct, but the image numbers will not match the prospective images.
- Only start Retro Recon when scanning is complete. Do not delete queued retros while scanning is active; scanning could stop.
- If maximum A-P or R-L offset is selected in Retro Recon, one image may fail to reconstruct. Select an offset that is 0.5mm less than the maximum value allowed.
- The start/end locations for a retro reconstruction may be different from Recon 1 and 2 and 3 when a PMR with a thicker slice than Recon1 is prescribed for the seies.
- Retro Recon may hang and core. If the Retor Recon screen fails to display or queue retrospective image, Retro Recon has cored. Reboot the system to recover.

RECON MANAGEMENT

- Scout queue entries can be selected in Delete Retro queue entries. Only select Scout queue entries to delete if you are sure you do not need the scout image reconstructed.
- Before saving or restoring scan data make sure the system is idle and no Archive, Network or Filming is active. No other features should be accessed until the save or restore is complete.
- Recon Management may hang while trying to display the menu. Finish the current exam if scanning and shutdown and restart the system to correct the problem.
- If you wish to cancel a Save or Restore scan data, the cancel button will cancel only after the current scan file is saved or restored. A helical scan file contains a large amount of data even though it is one scan file, that file can take up to 30 minutes or more to save. You are unable to scan patients while saving or restoring Scan Data.
Make sure that you have ample time to complete the save or restore before beginning.
- Scan data save or restore is active when the dialog message indicating saving or restoring scan file is displayed.
- When saving scan data, after the save is complete, select Restore to verify the data is stored on the disk.

DIRECT3D – PURCHASED OPTION

- Some error messages for Direct3D may refer to the feature as VAV. Initially, the feature was called Volume AutoView.
- Direc3D is not compatible with Biopsy, a tilted gantry or with multi-slice prospective reconstructions.
- If an invalid number for Zoom is entered, the system defaults the entry for the field to 0.1.
- When creating Screen Saves, wait for the image to re-render completely before pressing [Screen Save] or only the portion of the volume which was rendered will be captured in the Screen Save.

DIRECT3D CONTINUED

Filming is not available in AutoReview. There is a sequence of events where the Film button may appear to be available. Do not select Film as it may cause all filming to stop. The message "Film Composer not empty." will be displayed. A system shutdown may be required to recover.

- If Standard resolution is used for data sets with 256 or more images, the rendering will have corrupt data. Use Enhanced for large volumes (250-500 images).
- Direct3D rendering may have erroneous data at the edge of the volume for standard resolution renderings when clipping is used. To eliminate this data, turn clipping off.
- Advantage Windows 4.0 does not currently support the Direct3D Save State Series. The Save State Series for Direct3D will be supported in the next release for AW 4.1 software.

CT PERFUSION ON OC – PURCHASED OPTION

- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- The CT Perfusion Control Panel may be displayed on the current desktop if CT Perfusion is launched from the Browser and the desktop is changed before CT Perfusion screens are completely displayed.

AUTO TRANSFER

- A series may fail to Auto Transfer if [Next Series] then [Create New Series] is selected after a scout scan is prescribed, but not scanned, i.e.; pause and return to view edit is selected.

DISPLAY

- The following annotation changes will be seen when viewing LightSpeed QX/i images on a Advantage Windows version 1.2 or 2.0 Workstation,
 - Lung Algorithm will be listed as EXP-1
 - The product name will be displayed as a ?
- The Exam Rx List Select and Image Works browsers may in some cases not list a exam, series or image. This may be seen after an interruption of power to the system or the database may be updating slowly. The patient does not need to be rescanned.

Verify the following:

- Type the Exam, Series, Image on the accelerator line to display the image,
 - Select another exam in the browser this will cause the browser to update,
 - The browser may update after a slight delay,
 - Reboot the system,
 - A second reboot may be needed if a power fail has occurred.
- The browser may update slowly to reflect Screen Save images created in an exam. If the Screen Save images are not listed in the browser wait for them to appear. The time to appear may be up to 15 minutes if the Automatic Database check and recovery is in progress when the Screen Save images are created.
- In general, wait for a display action to complete before entering in another command.
- With the system configured in French or German language, if a comma is entered in explicit magnify, i.e. 2,3 the comma will not be interpreted as a decimal. The magnification factor is applied without the decimal. In the example shown, 2,3 the image would only be magnified 2X not 2.3X. Always enter a period to designate a decimal and have the correct magnification applied.
- Cross Reference lines will post incorrectly if sort by image location is selected. Make sure that sort by number is selected before posting Cross Reference lines.
- Cross Reference lines may not post with some combinations of DFOV and RAS center. RAS centers at the edge of the DFOV and small DFOV's will exhibit the problem. Retro reconstruct the images to a larger DFOV if posting of the cutlines is desired.
- Image annotation may overlap or not be fully shown if 32 character patient name is entered for 3D, Navigator, Reformat, MID formats, DentaScan images.
- Text and Series pages only display 24 characters for Patient Name.
- Text pages will not show foreign characters entered using the ALT GR key.
- ROI and STD deviation numbers may be reported as zero after a Zoom is applied to an image with a ROI posted. This is due to partial pixel contained in the ROI. Adjust the size of the ROI to include full pixels and recalculate the ROI.
- No text will be entered in the accelerator line if the the mouse cursor is over the film composer.
- Some combinations or E/S/I will fail to display an image, if cap locks is on. To select an exam series and image from the accelerator line use the following format when cap locks is on: For example to select Exam 576 Series 2 image 3 type E576 2 3 on the accelerator line.

DISPLAY CONTINUED

- Selection of an exam series or image may fail a second time after a exam series or image is displayed. If the desired exam series or image does not display then select the desired images from the browser.
- If the system is shut down with only one tic mark displayed,(ie vertical or horizontal), on reboot no tic marks will be displayed. Select [Image Works] [Viewer] [User Preferences] [Tic Marks on] to reset the tic mark display after every reboot.
- Image location seen on the scanner may differ from that seen on an AW 3.1 system for tilted images. LightSpeed reports the image location as iso-center of the image. AW 3.1 reports image location as the image center for the image. With tilted images there will be a difference in the location numbers propotional to the off center distance times the sin of the tilt angle. The image location can be reported as image center on LightSpeed by using the Report Cursor function.

EXAM RX DISPLAY

- Display may reset it's self after a software problem. You may see display stop or not accept your mouse input. Wait for a few seconds and display will automatically reset. You will have to redisplay the images you wish to work on.
- The MIROI pop up may display but will not function. This will occur if a Auto View or Cross reference viewport is in primary focus (blue border) MIROI is not valid in these viewport types. Place a Free or Auto Link viewport in primary focus and reselect the MIROI button to continue.
- Annotation in the AutoView viewport may appear black and shadowed. Move your cursor over the viewport to reset the annotation to white.
- Trackball paging will not start if paging has been started, (i.e. middle button has been doubled clicked) and no images have been reviewed, (i.e. the trackball was not used) and paging was cancelled. The series must be redisplayed from the browser for paging to start.
- Any Report Pixel or MIROI chart which is not screen saved, but is filmed will show the Exam Series and Image as 1000/1/1 on the film composer icon if E/S/I is selected. This is the exam/series/image number the system assigns to this chart display The patients exam number is listed correctly on the filmed image.
- Image selection from the accelerator line will not function if the primary viewport contains a MIROI plot or Report Pixels chart. Use List Select to display a new image.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be saved as defaults.
- ExamRx Display may hang. In some cases, switching to the Image Works desktop and back to ExamRx Display may clear the problem. If this doesn't work, perform a system shutdown and reboot the system.

IMAGE WORKS DISPLAY

- The Reference image display will **not** be filmed when F4 (Print Series) is used.
- WW and WL adjustments will only be maintained on individually selected images if the middle mouse button is used. If the accelerator line or Presets are used all images will update regardless of primary or secondary focus of the images.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be applied and not held if the viewer is closed. Use User Preferences in the viewer to save the settings as a default if desired.
- Reference Image may select the wrong scout series, i.e. not the scout series the images were prescribed from, if one or more scout series exists. If the wrong scout is selected place the Reference Image viewport in primary focus and enter the scout series number desired on the accelerator line.
- Window/Levels do not display stored value set in Auto Film when +/- series is used. The W/L used is the same as the last series displayed.
- Images created in Add/Sub are displayed with value of WW 4098 and WL 1024.
- If paging is selected while the system is in Compare Mode, the upper left viewport is left blank. Return to the Browser and select the desired series for paging and restart the Viewer.
- Accelerator Command Line entries for E/S/I do not act properly. The first image in the series is always displayed instead of the requested image within the series.
- Paging for multiple series does not work. Currently, only one series can be set into a paging loop. If additional series are selected in other viewports, they will not be paged.

EDIT PATIENT DATA

- The system may hang if Edit Patient is started while the following operations are in progress:
 - Network Receive
 - Prospective or Retrospective reconstruction is active
 - Archive Restore
 - Image from the exam is currently displayed in a viewport. Select the viewport and enter *blank* to clear the viewport.

To avoid this, confirm Edit Patient during idle times in network receive, active recon or archive restore.

- Patient age may be seen as zero in Edit Patient Data even though a birthdate is entered. Enter the birthdate again and patient age will be calculated.

REFORMAT

- When selecting reformat wait until the % loading dialog is displayed before switching to the Exam RX desktop. This will avoid having to select Load slices to start the reformat model build when returning back to the Image Works desktop.
- The reformat location will not be annotated on images filmed when Printer is selected in Batch Mode. If it is desired to have the reformat location annotated, then use Save in Batch Mode and film the images through Print Series.

- In Batch mode, images reconstructed using “Rotation” or “Oblique” modes will start with number 2 while images created using “First View/Last View” start with 1. In Rotation and Oblique modes, a plan of the reconstructions is screen saved and becomes image 1 for the series.
- Intermittently some images created in reformat will not archive. These series may contain corrupt information in the image file format. They will not be able to be archived. Delete them from the Archive queue and do not try to archive them again.

FILMING

- Images or Auto Film control buttons may fail to display in the AutoFilm viewport. This may occur after a switch between desktops. To display the images change to a different AutoView layout, then switch back to your desired layout.
- The Reference image display will not be filmed when F4 (Print Series) is used.
- Keep current on Auto Film.
- Any of the Auto Film control buttons may be activated when the cursor is over the button and the space bar is selected. Place the cursor in the Auto Film viewport only when needed to avoid this problem.
- Auto Film may fail to film. This may occur when the Auto Film viewport selection shows active and the Auto Film status shows paused. Toggle the [Pause] button in the Auto Film viewport and select [Start New Sheet] or [Continue Same Sheet] to restart filming.
- When scanning multiple one rotation Axial groups, images are filmed out of order when Auto Film is selected. Turn Auto Film off for these scan groups.
- Auto Film may fail to display images if recon is having trouble reconstructing the image or if images have failed to install in the data base. This may also occur if a large exam has not reconstructed all images and scanning is started on a new exam.
- The following dialog will be posted if Auto Film cannot display images.

➤ Exam 100
 ➤ Series 2
 ➤ Cannot find 10 images

Skip Missing images
Continue
Cancel Film Series

You are presented with 3 choices:

1. Skip Missing images will skip the images which Auto Film cannot find. For example if Auto Film had filmed images 1-10 and you then got the message that 10 images were missing and selected Skip Missing Images, Auto Film would then start filming again with image 21.
2. Continue will look for the images again, if they are not found then the dialog will be posted again. Before selecting continue verify that recon is active, that the images have been reconstructed, the missing images are not suspended or paused in the Recon Queue and the images are able to be displayed. If the images are suspended or paused in the recon queue, then unsuspend or restart recon to reconstruct the images before selecting Continue.
3. Cancel Film Series will cancel Auto Film for the series currently being Auto Filmed. The images will then need to be manually filmed.

FILMING CONTINUED

- If you get this dialog because you have begun an exam while the previous exam still has images to reconstruct, then Select Continue and select Pause Auto Film, resume Auto Film when the Exam Series and Images from the previous exam have reconstructed.
- When prescribing Mag factor for auto film, scout images if the scout is longer than 500 mm then use a magnification factor less than 1 to display the entire scout.
- The manual film composer may display when confirm is selected for scanning if the composer had been closed by selecting iconify in the upper right corner of the composer.
- It's best to run Print Series from only one desktop at a time.
- It's best to resolve any paused queue entry as soon as possible.
- Print Series may pause automatically under heavy system load.
- If Anonymous Patient is selected for an exam where Auto Film is in progress, the system may fail to cross reference the slices on the scout. If Anonymous Patient is required for series which are still active in Auto Film, wait till Auto Film is complete for the series before proceeding with Anonymous Patient.
- To install a SMPTE pattern select the following. Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. and will be listed as Exam 1000. The patient Name will be listed as SMPTE.
 - Select Service Desktop
 - Select [Diagnostics]
 - Select [Display Processing]
 - Select [Install SMPTE Image]
- Scout scans for an exam can be deleted while Auto Film is still active for the exam. Do not delete scout images for an exam until Auto Film is completed for the exam.
- Full Annotation instead of partial annotation will be filmed when using F3 (Film MID) in the Image Works desktop when full annotation is selected.
- The format built in a protocol may be changed when the protocol is used and a message that the format was changed due to invalid format, even if the format was valid. Verify in Auto Film set up, the desired format is selected before continuing to scan.
- Under the following conditions, images auto filmed from 2 or more scan groups in multiple image display (MID) will not carry over an applied enhancement filter.
 - Frame format is 2 on 1 or 4 on 1 in a single viewport.
 - Enhancement filter is selected.
 - First group has a number of images that leaves blank frames.
 - The images displayed in the previous blank frame for the next group will not have the filter applied.

ARCHIVE

- The feature status area or the browser may report an Exam has saved even though all images have not been saved to MOD. Scroll through the images in the image window. Highlight the images listed as Archive N and resave them.
- Restoring Exam, Series or images that already exist on the system disk will not post a message that the images are restored or that they already exist. If you have restored images, but get no message that they have been restored, verify that they don't already exist on the system disk.
- When the archive media gets close to being full, the system will always look to see if the exam, series or image can fit on the MOD, if it is desired to have exams sequentially on a MOD, place a new MOD in the drive when first notified the disk may be full.
- If Label is selected in Archive, finish the Label process, don't select cancel on the Label pop up. Archive will be non functional, a system reboot will be necessary to recover.
- It's best to resolve any paused queue entry as soon as possible.
- To minimize corruption of MODs, it is extremely important the MOD media be detached and removed from the MOD drive before rebooting the system.
- When restoring from a MOD, it is recommended to write protect the media before placing it in the drive. If the system is unable to read the media, detach and remove the write protect and see if the system is able to perform a recovery of the media to access the data.

NETWORK

- LightSpeed images will **not** transfer if Advantage Net protocol is used. Always use DICOM protocol to send LightSpeed images
- LightSpeed images **cannot** be sent to a HiLight Advantage, HiSpeed Advantage or CT Independent console. These systems do not support DICOM receive.
- Some 3rd party workstations may fail to receive LightSpeed scout images. This is due to the matrix size of the scout image. These stations do not support receiving matrix sizes greater than 512. Some LightSpeed scouts images have matrix sizes greater than 512. If it is desired to have the scout image on the workstation, Screen Save the scout and then transfer the screen save image to the workstation.
- If you query a LightSpeed system from a CT/i or Advantage Windows system in Advantage Net, only Advantage format image exams will be displayed, No LightSpeed exams will be shown. Always query a LightSpeed system using DICOM protocol.
- Lateral scouts displayed on an AW 3.1 system will initially display with zero rotation; they should display with a 270-degree rotation. Rotate the image in the left direction using the rl command on the command line to display the scout in the desired format.
- Images networked to a Advantage Windows 3.1 workstation running software versions prior to 3.1_07 will display a DFOV less than what is displayed on the scanner. This is due to the AW not taking in account the pixels under the focus border of the viewport.

NETWORK CONTINUED

- Images networked to a AW 3.1 will have the following annotation missing:
 - Accession number,
 - Date of Birth,
 - 4i, 2i or 1i recon mode on Axial, Retro Axial or Axial Reformatted images,
 - Series type on Retro images,
 - Table Speed on Helical images,
 - Images are annotated +C when Oral Contrast is used. View the Series Text Page to see if IV contrast was used.
- The number of images indicated in a series may be incorrect on the remote browser when a query is made from a AW or CT/i station.
- Series types may be listed differently when images are networked to an Advantage Windows 1.2, 2.0 or 3.1 system than that shown on the LightSpeed system
- Exams with Swedish, German or French characters in any Patient Info field will not be transferred to a Advantage Windows.
- When using DICOM protocol, the entered host name must match in spelling and case sensitive, otherwise a connection error message will be displayed.
- It's best to resolve any paused queue entry as soon as possible.

IMAGE MANAGEMENT

- Don't let image space fall below 200 images on a single disk system (20,000-image storage). This will ensure there is room on the system disk to confirm scans and to install reconstructed images.
- Remove images when scanning is idle. This will assure that all images are reconstructed, displayed and installed into the database.
- Scout scans for an exam can be deleted while Auto Film is still active for the exam. Do not delete scout images for an exam until Auto Film is completed for the exam.

SHERLOCK CBT

- To access information about CTDI Select Global Index from the main menu then Select CTDIw.
- The index feature works only within a module.
- When setting a bookmark don't put spaces in the name you enter. If spaces are in the name the bookmark will not be found and the following message will be posted "Sorry Wrong CD-Rom"
- Find will list all bookmarks created on your system. These book marks may come from more than one CBT disk. If you select a bookmark that is for a disk which you currently are not viewing a message will be posted "Wrong CD-Rom." When naming bookmarks enter a character to identify which CBT disk. the bookmark belongs to.

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